

Book V of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Chapter 8

Proposition V.8

8.1 Introduction

Proposition V.8 is the first proposition in the present reconstruction of Book V where the Eudoxean comparison of ratios requires a genuinely Archimedean argument. The statement is elementary to read but structurally important: if one positive magnitude is greater than another, then this strict inequality is detected by a suitable pair of positive multiples in the sense of Euclid’s Definition 7.

The proposition is also the first substantial consumer of the general magnitude layer developed in `HilbertInterfaceV.lean`. The formal proof uses strict order on positive segment classes, positive differences, iterated addition, Archimedes’ principle, and a least-exceeding-multiple construction.

The production file is

```
CGJteamLab/Proposition5_8.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

8.2 Euclid’s Statement

Euclid states Proposition V.8 as follows:

Of unequal magnitudes, the greater has to the same a greater ratio than the less has; and the same has to the less a greater ratio than it has to the greater.

In the present positive-magnitude notation, if $c < a$, the proposition has two parts:

$$a : d > c : d,$$

and

$$d : c > d : a.$$

The first compares two ratios with the same second term. The second compares two ratios with the same first term.

8.3 Magnitude Configuration

There is no planar or spatial diagram in V.8. The configuration consists of positive magnitudes and their positive integer multiples.

For the first half, assume

$$c < a.$$

By positive difference there exists a positive magnitude e such that

$$c + e = a.$$

The formal proof then chooses a multiple of e large enough to exceed the common second magnitude d :

$$d < M_m(e).$$

It next chooses the least n such that

$$M_m(c) < M_n(d).$$

The comparison at this least index is the decisive Eudoxean witness.

For the second half, the same structure is used after writing

$$c + e = a$$

with the roles of first and second terms interchanged.

8.4 Eudoxus Definition 5 and Definition 7

The reconstruction distinguishes two notions.

8.4.1 Equality of ratios

The relation

```
HilbertEudoxusProportion Geo a b c d
```

formalizes Euclid's Definition 5. For every pair of positive natural multiple indices m, n , the three possible comparisons

$$M_m(a) < M_n(b), \quad M_m(a) = M_n(b), \quad M_n(b) < M_m(a)$$

must agree with the corresponding comparisons between c and d .

8.4.2 Greater ratio

The relation

```
HilbertEudoxusGreaterRatio Geo a b c d
```

formalizes the strict comparison of ratios used in V.8. It requires positive-multiple indices m, n such that

$$M_n(b) < M_m(a)$$

while

$$\neg(M_n(d) < M_m(c)).$$

The negation in the second clause is essential. Equality on the $c : d$ side is allowed; replacing this clause by the strict opposite inequality would be stronger than the Eudoxean definition used here.

8.5 Positive Magnitude Infrastructure

The objects of the proof are

```
HilbertPositiveSegmentClass Geo
```

rather than raw point pairs. A positive segment class is a congruence class of non-null Hilbert segments.

The strict order is

```
HilbertPositiveSegmentLess Geo a b
```

and is induced from the geometric strict segment order.

The proof also uses positive addition and positive natural multiples:

```
hilbertPositiveSegmentMultiple Geo n a
```

with shifted indexing:

$$M_0(a) = a, \quad M_1(a) = a + a, \quad M_2(a) = a + a + a, \quad \dots$$

Thus every natural index already denotes a positive multiple; there is no zero magnitude in the type.

8.6 The Archimedean Step

The central infrastructure theorem is

```
hilbertPositiveSegment_archimedean
```

which states that for positive magnitudes a, b there exists m such that

$$b < M_m(a).$$

This theorem is not an additional axiom of Book V. It is derived in `HilbertInterfaceV.lean` from the existing geometric Archimedean structure

```
HilbertArchimedeanPlane
```

after normalizing the weak Hilbert segment chain to positive segment classes.

For V.8 the theorem is applied to the positive remainder e . In the first half it gives

$$d < M_m(e).$$

This is the step that makes it possible to separate the two ratios by finite multiples.

8.7 Least Exceeding Multiple

Archimedes gives existence of an exceeding multiple, but V.8 also needs minimality.

The reusable theorem is

```
hilbertPositiveSegment_exists_least_exceeding_multiple
```

and produces a least n such that

$$x < M_n(a),$$

together with

$$k < n \implies \neg(x < M_k(a)).$$

For the first half of V.8 we apply it with

$$x = M_m(c), \quad a = d.$$

Hence

$$M_m(c) < M_n(d),$$

and no earlier multiple of d already exceeds $M_m(c)$.

This least-index condition is the formal replacement for the implicit “first multiple beyond” step in the classical magnitude argument.

8.8 Euclid's Classical Proof Pattern

The first half of the proposition can be organized as follows.

Assume

$$c < a.$$

Take the positive difference e with

$$a = c + e.$$

Choose m such that

$$d < M_m(e).$$

Choose the least n such that

$$M_m(c) < M_n(d).$$

By minimality, the preceding multiple does not exceed $M_m(c)$. Consequently the next step of size d , together with the inequality $d < M_m(e)$, forces

$$M_n(d) < M_m(c) + M_m(e).$$

Since positive multiplication distributes over positive addition,

$$M_m(c) + M_m(e) = M_m(c + e) = M_m(a).$$

Therefore

$$M_n(d) < M_m(a).$$

At the same time,

$$M_m(c) < M_n(d),$$

so by asymmetry

$$\neg(M_n(d) < M_m(c)).$$

The pair m, n therefore witnesses

$$a : d > c : d.$$

The second half is the dual argument with the common first term fixed.

8.9 Classical Dependency Pattern

The mathematical dependency pattern used in the reconstruction is

$$\begin{array}{c}
 c < a \\
 \Downarrow \\
 \exists e > 0, \ c + e = a \\
 \Downarrow \\
 \text{Archimedes: choose } m \text{ with } d < M_m(e) \\
 \Downarrow \\
 \text{least } n \text{ with } M_m(c) < M_n(d) \\
 \Downarrow \\
 M_n(d) < M_m(a) \\
 \Downarrow \\
 \neg(M_n(d) < M_m(c)) \\
 \Downarrow \\
 a : d > c : d.
 \end{array}$$

The second clause of V.8 follows from the same architecture after interchanging the two ratio positions.

8.10 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_8
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]

```

```

(a c d : HilbertPositiveSegmentClass Geo)
(hca : HilbertPositiveSegmentLess Geo c a) :
HilbertEudoxusGreaterRatio Geo a d c d /\
HilbertEudoxusGreaterRatio Geo d c d a := by

constructor
case left =>
  exact
    hilbertEudoxus_v8_greater_first_same_second
      Geo a c d hca
case right =>
  exact
    hilbertEudoxus_v8_same_first_lesser_second
      Geo d c a hca

```

The final Euclid-numbered theorem is intentionally short. The substantive proof is carried by the two proposition-local lemmas:

```

hilbertEudoxus_v8_greater_first_same_second
hilbertEudoxus_v8_same_first_lesser_second

```

which formalize the two halves of Euclid’s statement.

8.11 Lean Proof Architecture: First Half

The first proposition-local theorem is

```

hilbertEudoxus_v8_greater_first_same_second

```

with target

$$c < a \implies a : d > c : d.$$

The proof proceeds through the following data.

First,

```

hilbertPositiveSegment_exists_add_of_less

```

produces e with

$$c + e = a.$$

Second,

```

hilbertPositiveSegment_archimedean

```

produces m with

$$d < M_m(e).$$

Third,

```
hilbertPositiveSegment_exists_least_exceeding_multiple
```

produces the least n such that

$$M_m(c) < M_n(d).$$

The proof then splits on n .

8.11.1 Case $n = 0$

Here

$$M_0(d) = d.$$

Since

$$d < M_m(e)$$

and

$$M_m(e) < M_m(c) + M_m(e),$$

we obtain

$$d < M_m(c) + M_m(e) = M_m(a).$$

The least-index inequality itself gives

$$M_m(c) < d,$$

hence asymmetry supplies the required negated comparison.

8.11.2 Case $n = k + 1$

Minimality gives

$$\neg(M_m(c) < M_k(d)).$$

By trichotomy either

$$M_k(d) < M_m(c)$$

or

$$M_k(d) = M_m(c).$$

The reverse strict inequality is impossible by minimality.

Combining either remaining case with

$$d < M_m(e)$$

gives

$$M_k(d) + d < M_m(c) + M_m(e).$$

Using the recursion for positive multiples and distributivity,

$$M_{k+1}(d) < M_m(a).$$

Thus m, n satisfy Definition 7.

8.12 Lean Proof Architecture: Second Half

The second proposition-local theorem is

```
hilbertEudoxus_v8_same_first_lesser_second
```

with target

$$b < d \implies a : b > a : d.$$

The proof is dual.

Write

$$b + e = d.$$

Choose n such that

$$a < M_n(e),$$

then choose the least m such that

$$M_n(b) < M_m(a).$$

The minimality analysis of m shows

$$M_m(a) < M_n(d).$$

Hence asymmetry gives

$$\neg(M_n(d) < M_m(a)),$$

while the least-exceeding construction already gives

$$M_n(b) < M_m(a).$$

These are exactly the two clauses required to witness

$$a : b > a : d.$$

8.13 Hidden Formal Data

Several details that are implicit in Euclid’s magnitude language must be explicit in Lean.

8.13.1 All magnitudes are positive

The type

```
HilbertPositiveSegmentClass Geo
```

contains only non-null segment classes. Therefore subtraction is not a primitive total operation. The positive difference e is obtained only under a strict inequality.

8.13.2 Positive multiples use shifted indices

The index 0 denotes one copy, not zero copies. This is why the least exceeding multiple proof has a genuine base case $n = 0$.

8.13.3 Minimality is explicit

The phrase “take the first multiple which exceeds” is formalized as an existence theorem over $\mathbb{N}$ with a universal minimality clause.

8.13.4 The greater-ratio witness is existential

V.8 does not prove a numerical quotient inequality. It produces the multiple indices m, n required by Eudoxus Definition 7.

8.14 Relationship with Earlier and Later Book V Propositions

In the present reconstruction V.8 is the first Euclid-numbered theorem in Book V that uses the full Archimedean magnitude infrastructure.

Its main later consumers are V.14 and, through V.14, V.16.

For equal ratios

$$a : b = c : d,$$

V.14 needs to transport strict comparison from a, c to b, d . The strict part begins with V.8: from $a < c$ one obtains a strict comparison of ratios with a common second term. V.13 transports that ratio comparison through the given equality, and V.10 converts it back to an inequality of magnitudes.

Thus the key dependency chain is

$$V.8 \longrightarrow V.13 \longrightarrow V.10 \longrightarrow V.14 \longrightarrow V.16.$$

This is also why V.8 was the first substantial theorem formalized when the Book V layer was introduced for the later reconstruction of XI.23.

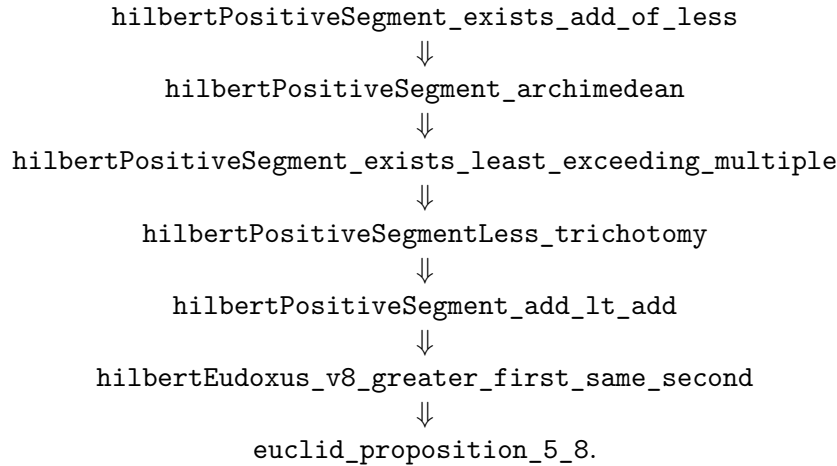
8.15 Classical DAG versus Formal DAG

The formal proof keeps the classical Eudoxean structure but makes several implicit steps explicit.

Magnitude-level DAG.

$$\begin{array}{c}
 c < a \\
 \Downarrow \\
 c + e = a \\
 \Downarrow \\
 d < M_m(e) \\
 \Downarrow \\
 \text{least } n : M_m(c) < M_n(d) \\
 \Downarrow \\
 M_n(d) < M_m(c) + M_m(e) \\
 \Downarrow \\
 M_n(d) < M_m(a) \\
 \Downarrow \\
 a : d > c : d.
 \end{array}$$

Lean-level DAG.



The second half has the same architecture with the ratio positions interchanged.

8.16 Structural Lesson

V.8 shows why Book V cannot be reduced to a few symbolic ratio identities. The strict comparison of ratios depends on the order theory of magnitudes and on an Archimedean finite-multiple principle.

The formal reconstruction makes this especially clear. No real numbers, coordinates, lengths as scalars, or division operation are introduced. Everything is expressed synthetically in terms of:

- congruence classes of positive segments;
- strict segment order;
- positive addition;
- iterated addition;
- Archimedean existence of a sufficiently large multiple;
- well-ordering of the natural-number multiple index.

This is precisely the kind of infrastructure needed for a source-faithful Eudoxean development.

8.17 Conclusion

Proposition V.8 is the first major order theorem of the reconstructed Eudoxean ratio calculus.

Starting from a strict inequality of positive magnitudes, the proof constructs finite multiples that witness a strict inequality of ratios in the exact sense of Definition 7. The decisive ingredients are the positive remainder, Archimedes' principle, and the least multiple crossing a prescribed magnitude.

The final Euclid-numbered theorem is a short wrapper, but the mathematical content is substantial. It is the gateway from the additive ordered magnitude layer to the later comparison theorems

V.10, V.13, V.14, and ultimately to V.16 alternando, which is required in the classical route to Euclid XI.23.

Chapter 10

Proposition V.10

10.1 Introduction

Proposition V.10 is the order-reflection companion to Proposition V.8. Where V.8 starts with a strict inequality of magnitudes and constructs a strict inequality of ratios, V.10 starts with a strict ratio comparison and recovers the underlying strict inequality of magnitudes.

In the present reconstruction this converse direction is substantially shorter than V.8. No Archimedean argument and no least-exceeding-multiple construction are needed. The proof simply unpacks the witness supplied by Eudoxus' greater-ratio relation, applies trichotomy to the candidate magnitudes, and excludes the two wrong comparison cases.

The production file is

```
CGJteamLab/Proposition5_10.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

10.2 Formal Reading of Proposition V.10

The production file packages two comparison forms.

The first says:

$$a : d > c : d \implies c < a.$$

The ratios have the same second term d .

The second says:

$$a : b > a : d \implies b < d.$$

The ratios have the same first term a .

These are exactly the two forms needed later in the reconstruction of V.14.

10.3 The Greater-Ratio Witness

Recall that

HilbertEudoxusGreaterRatio Geo a b c d

means that there exist positive-multiple indices m, n such that

$$M_n(b) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(c)).$$

Thus a strict comparison of ratios is not represented by a scalar quotient. It is represented by a concrete pair of finite multiples that separates the two ratios.

V.10 shows that when one term is shared, such a witness forces the expected order of the remaining magnitudes.

10.4 First Comparison Form

Assume

$$a : d > c : d.$$

Unpacking the greater-ratio witness gives m, n with

$$M_n(d) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(c)).$$

Now apply trichotomy to c and a .

There are three cases:

$$c < a, \quad c = a, \quad a < c.$$

The first case is the desired conclusion.

If

$$c = a,$$

then the two multiple comparisons in the witness become the same proposition. The witness would simultaneously assert

$$M_n(d) < M_m(a)$$

and its negation. This is impossible.

If instead

$$a < c,$$

then positive multiplication preserves strict order:

$$M_m(a) < M_m(c).$$

Together with

$$M_n(d) < M_m(a),$$

transitivity yields

$$M_n(d) < M_m(c),$$

contradicting the negative clause of the greater-ratio witness.

Therefore only

$$c < a$$

remains.

10.5 Second Comparison Form

Assume

$$a : b > a : d.$$

The witness gives

$$M_n(b) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(a)).$$

Apply trichotomy to b and d .

If

$$b < d,$$

we are done.

If

$$b = d,$$

the positive and negative witness clauses become contradictory.

If

$$d < b,$$

positive multiplication gives

$$M_n(d) < M_n(b).$$

Combining this with

$$M_n(b) < M_m(a)$$

gives

$$M_n(d) < M_m(a),$$

again contradicting the negative witness clause.

Hence

$$b < d.$$

10.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_5_10
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d : HilbertPositiveSegmentClass Geo) :
  (HilbertEudoxusGreaterRatio Geo a d c d ->
```

```

      HilbertPositiveSegmentLess Geo c a) /\
    (HilbertEudoxusGreaterRatio Geo a b a d ->
      HilbertPositiveSegmentLess Geo b d) := by

constructor
case left =>
  exact
    hilbertEudoxus_v10_same_second_lt_first
      Geo a c d
case right =>
  exact
    hilbertEudoxus_v10_same_first_lt_second
      Geo a b d

```

As in V.8, the final Euclid-numbered theorem is a thin wrapper over two proposition-local lemmas:

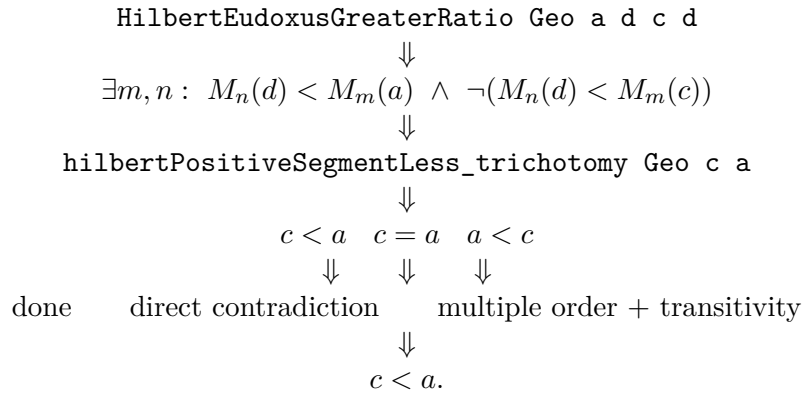
```

hilbertEudoxus_v10_same_second_lt_first
hilbertEudoxus_v10_same_first_lt_second

```

10.7 Lean Proof Architecture

The first half has the following formal DAG:



The second half is identical in structure, with trichotomy applied to b, d .

10.8 Infrastructure Used

Unlike V.8, the proof does not use

```

HilbertArchimedeanPlane
hilbertPositiveSegment_archimedean
hilbertPositiveSegment_exists_least_exceeding_multiple

```

at all.

The essential reusable infrastructure is only:

```
hilbertPositiveSegmentLess_trichotomy
hilbertPositiveSegmentMultiple_lt
hilbertPositiveSegmentLess_trans
```

together with the definition of

```
HilbertEudoxusGreaterRatio
```

itself.

This is mathematically significant: V.10 is an order-theoretic consequence of the Eudoxean witness once the existence of that witness is assumed.

10.9 Relationship with V.8

V.8 and V.10 form a natural pair.

V.8 gives, in the common-second-term form,

$$c < a \implies a : d > c : d.$$

V.10 gives the converse:

$$a : d > c : d \implies c < a.$$

Similarly, for a common first term,

$$b < d \implies a : b > a : d$$

comes from V.8, while

$$a : b > a : d \implies b < d$$

comes from V.10.

Thus the two propositions together identify strict order of magnitudes with the corresponding strict ratio comparison in these common-term configurations.

10.10 Role in Proposition V.14

V.10 is used later as the final step in the strict comparison part of V.14.

Suppose

$$a : b = c : d$$

and

$$a < c.$$

V.8 first gives a strict ratio comparison with a common second term. V.13 transports that comparison through the equality of ratios. V.10 then converts the resulting ratio comparison back into

$$b < d.$$

The dependency chain is therefore

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10} \longrightarrow \text{V.14}.$$

10.11 Hidden Formal Data

Two features of the Lean proof deserve attention.

10.11.1 The witness indices are retained

The proof does not discard the particular m, n supplied by the greater-ratio relation. Those same indices are used to derive the contradiction in the wrong trichotomy case.

10.11.2 Order preservation under a fixed positive multiple

The crucial move in the wrong strict-order case is

$$a < c \implies M_m(a) < M_m(c),$$

or, in the second half,

$$d < b \implies M_n(d) < M_n(b).$$

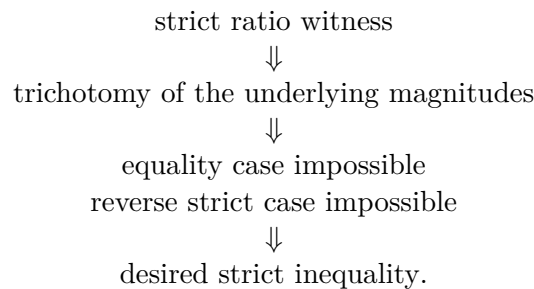
This is the role of

```
hilbertPositiveSegmentMultiple_lt
```

and explains why the positive-multiple order theory belongs in the reusable Book V interface rather than in a proposition-specific file.

10.12 Classical DAG versus Formal DAG

The conceptual proof is extremely short:



The Lean proof follows exactly this structure.

There is no auxiliary geometric construction and no proposition-specific magnitude object. The extra formal detail consists only in spelling out the multiple indices and the transitivity step that produces the contradiction.

10.13 Structural Lesson

V.10 separates two logically distinct layers of the Book V development.

The difficult existence problem belongs to V.8: one must construct suitable multiple indices witnessing a strict ratio comparison.

Once such indices exist, V.10 is almost purely order logic. It demonstrates that the Eudoxean Definition 7 is strong enough to reflect strict magnitude order in the presence of a common ratio term.

This division is useful for the later architecture. V.14 does not need to repeat any Archimedean reasoning: it uses V.8 to manufacture a ratio comparison, V.13 to transport it, and V.10 to read the resulting comparison back as a magnitude inequality.

10.14 Conclusion

Proposition V.10 is the order-reflection theorem of the reconstructed Eudoxean ratio calculus.

Its proof is short because all existence content has already been packaged in the definition of a greater ratio. By trichotomy, preservation of strict order under positive multiplication, and transitivity, the two wrong magnitude comparisons are excluded.

The result is the exact converse mechanism needed after V.8 and is a central input to V.14.

Chapter 11

Proposition V.11

11.1 Introduction

Proposition V.11 is structurally one of the simplest propositions in the present Book V reconstruction. Its statement is the transitivity of equality of ratios:

$$a : b = c : d, \quad c : d = e : f \implies a : b = e : f.$$

In the Eudoxean interface adopted by the project, equality of ratios is defined pointwise by preservation of all three comparisons between positive integer multiples. Consequently V.11 becomes direct composition of equivalences.

The production file is

```
CGJteamLab/Proposition5_11.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the Book V infrastructure.

11.2 Euclid's Statement

The proposition is represented in the production file by the statement:

Ratios equal to the same ratio are equal to one another.

Formally, if

$$a : b = c : d$$

and

$$c : d = e : f,$$

then

$$a : b = e : f.$$

No strict ratio comparison is involved.

11.3 Eudoxean Equality of Ratios

The underlying relation is

HilbertEudoxusProportion Geo a b c d

which says that for every $m, n \in \mathbb{N}$, using the project's shifted positive-multiple indexing, the three comparisons

$$M_m(a) < M_n(b),$$

$$M_m(a) = M_n(b),$$

and

$$M_n(b) < M_m(a)$$

are respectively equivalent to

$$M_m(c) < M_n(d),$$

$$M_m(c) = M_n(d),$$

and

$$M_n(d) < M_m(c).$$

Thus equality of ratios is literally a family of three equivalences indexed by all positive finite multiples.

11.4 Why Transitivity Is Immediate

Assume

$$a : b = c : d$$

and

$$c : d = e : f.$$

Fix arbitrary m, n .

For the less-than comparison we have

$$M_m(a) < M_n(b) \iff M_m(c) < M_n(d)$$

from the first proportion, and

$$M_m(c) < M_n(d) \iff M_m(e) < M_n(f)$$

from the second.

Composing the two equivalences gives

$$M_m(a) < M_n(b) \iff M_m(e) < M_n(f).$$

Exactly the same argument applies to equality:

$$M_m(a) = M_n(b) \iff M_m(e) = M_n(f),$$

and to the reverse strict comparison:

$$M_n(b) < M_m(a) \iff M_n(f) < M_m(e).$$

Since m, n were arbitrary, the three clauses are precisely Definition 5 for

$$a : b = e : f.$$

11.5 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_11
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d e f : HilbertPositiveSegmentClass Geo)
  (h1 : HilbertEudoxusProportion Geo a b c d)
  (h2 : HilbertEudoxusProportion Geo c d e f) :
  HilbertEudoxusProportion Geo a b e f := by
exact

```

```
hilbertEudoxusProportion_trans
  Geo a b c d e f h1 h2
```

The Euclid-numbered proposition is intentionally a one-step wrapper around the general interface theorem

```
hilbertEudoxusProportion_trans
```

already proved in `HilbertInterfaceV.lean`.

11.6 The Interface Theorem

The reusable theorem has the same mathematical content:

```
hilbertEudoxusProportion_trans
```

Its proof introduces arbitrary m, n , obtains the three comparison equivalences from each hypothesis, and composes them with

```
Iff.trans
```

for each comparison type.

There is no Archimedean step, no order trichotomy, and no construction of a greater-ratio witness.

11.7 Lean Proof Architecture

The complete formal DAG is

$$\begin{array}{c}
 h1 : \text{HilbertEudoxusProportion Geo a b c d} \\
 h2 : \text{HilbertEudoxusProportion Geo c d e f} \\
 \Downarrow \\
 \forall m, n, \text{ three Iff clauses from } h1 \\
 \forall m, n, \text{ three Iff clauses from } h2 \\
 \Downarrow \\
 \text{Iff.trans on each clause} \\
 \Downarrow \\
 \text{HilbertEudoxusProportion Geo a b e f} \\
 \Downarrow \\
 \text{euclid_proposition_5_11.}
 \end{array}$$

The proposition therefore contains no proposition-local helper lemmas.

11.8 Axiomatic Strength

The theorem requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

because these structures are already needed to form positive segment classes and their comparison relation.

In particular, V.11 does not require

```
HilbertArchimedeanPlane
```

and does not use continuity.

This is in sharp contrast with V.8, where the Archimedean principle is essential to manufacture a strict ratio witness.

11.9 No New Magnitude Construction

V.11 introduces no new magnitude, no positive remainder, and no finite multiple chosen by an existence theorem.

All objects

$$a, b, c, d, e, f$$

are supplied as hypotheses, and the conclusion is obtained solely by the logical structure of Definition 5.

Thus the proposition is an especially clear example of a theorem that becomes short after the correct reusable notion has been isolated.

11.10 Relationship with Definition 5

The brevity of the formal proof is not accidental. It reflects the design of

```
HilbertEudoxusProportion
```

itself.

Definition 5 was encoded as preservation of all three possible comparisons for all positive multiple pairs. Such a relation is automatically transitive because equivalence is transitive.

Accordingly, V.11 is not an additional arithmetic property imposed on ratios. It is a structural theorem of the chosen Eudoxean equality relation.

11.11 Relationship with Proposition V.16

V.11 is used directly in the formal proof of V.16 alternando.

The V.16 proof first uses the common-multiple content of V.15 to obtain

$$a : b = M_m(a) : M_m(b)$$

and

$$c : d = M_n(c) : M_n(d).$$

Starting from

$$a : b = c : d,$$

V.11 then composes these equalities of ratios to produce

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.14 can then transport the comparison between the first terms to the second terms. Since m, n are arbitrary, this yields the alternated proportion.

Hence the relevant dependency chain is

$$\text{V.15} \longrightarrow \text{V.11} \longrightarrow \text{V.14} \longrightarrow \text{V.16}.$$

11.12 Classical DAG versus Formal DAG

At the mathematical level:

$$\begin{array}{c} a : b = c : d \\ c : d = e : f \\ \Downarrow \\ a : b = e : f. \end{array}$$

At the Definition 5 level:

$$\begin{array}{c} \text{comparison of multiples for } a : b \\ \Updownarrow \\ \text{comparison of multiples for } c : d \\ \Updownarrow \\ \text{comparison of multiples for } e : f \\ \Downarrow \\ \text{comparison of multiples for } a : b \Updownarrow \text{comparison of multiples for } e : f. \end{array}$$

At the Lean level this is simply three uses of `Iff.trans` under universal quantification.

11.13 Structural Lesson

V.11 illustrates an important formalization principle.

A proposition can be historically central and still have a very short Lean proof when its mathematical content has already been placed at the correct abstraction level.

Here the work lies in defining Eudoxean equality of ratios correctly. Once that definition is available, transitivity is immediate.

The Euclid-numbered theorem should therefore remain a thin wrapper rather than duplicating the general proof in a proposition-specific file.

11.14 Conclusion

Proposition V.11 is the transitivity theorem for Eudoxean equality of ratios.

In the present architecture it is exactly the reusable theorem

`hilbertEudoxusProportion_trans,`

exposed under Euclid’s proposition number.

The proof requires no Archimedean reasoning and no construction. It consists only of composing the three comparison equivalences that constitute Definition 5. Its main later role is in V.16, where it composes common-multiple proportions with an existing ratio equality before V.14 performs the decisive comparison transfer.

Chapter 13

Proposition V.13

13.1 Introduction

Proposition V.13 is a transport theorem for strict ratio comparison. It combines equality of ratios, expressed by Eudoxus' Definition 5, with the strict greater-ratio relation represented by a witness in Definition 7.

The proof is short, but its role in the Book V dependency graph is important. It is the bridge that allows a strict comparison manufactured by V.8 to be moved through an existing equality of ratios before V.10 converts the transported ratio comparison back into an inequality of magnitudes.

The production file is

```
CGJteamLab/Proposition5_13.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

13.2 Formal Statement of the Comparison Pattern

Assume

$$a : b = c : d$$

and

$$c : d > e : f.$$

Then V.13 asserts

$$a : b > e : f.$$

The first hypothesis is an equality of ratios in the sense of Definition 5. The second is a strict ratio comparison in the sense of Definition 7.

The proposition therefore transports an already existing strict witness through an equality of ratios.

13.3 The Definition 7 Witness

The hypothesis

$$c : d > e : f$$

means that there exist positive-multiple indices m, n such that

$$M_n(d) < M_m(c)$$

and

$$\neg(M_n(f) < M_m(e)).$$

The crucial observation is that the second clause already has exactly the form needed for the target

$$a : b > e : f.$$

Only the first clause must be transported.

13.4 Transport through Definition 5

From

$$a : b = c : d,$$

Definition 5 gives, for the same indices m, n ,

$$M_n(b) < M_m(a) \iff M_n(d) < M_m(c).$$

The witness hypothesis contains the right-hand side:

$$M_n(d) < M_m(c).$$

Therefore the equivalence immediately yields

$$M_n(b) < M_m(a).$$

Together with the unchanged negative clause

$$\neg(M_n(f) < M_m(e)),$$

the same pair m, n witnesses

$$a : b > e : f.$$

No new multiple indices are constructed.

13.5 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_13
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d e f : HilbertPositiveSegmentClass Geo)
  (hProp :
    HilbertEudoxusProportion Geo a b c d)
  (hGreater :
    HilbertEudoxusGreaterRatio Geo c d e f) :
  HilbertEudoxusGreaterRatio Geo a b e f := by

rcases hGreater with
  \langle m, n, hCD, hNotEF\rangle

refine
  \langle m, n, ?_, hNotEF\rangle

exact
  (hilbertEudoxusProportion_gt_iff
    Geo
    a b c d
    hProp
    m n).2
  hCD

```

The entire mathematical proof consists of reusing the same witness m, n and transporting one strict comparison.

13.6 The Extraction Lemma

The reusable interface lemma is

```

hilbertEudoxusProportion_gt_iff

```

which extracts the reverse-strict-comparison clause of Definition 5.

For fixed m, n and a proportion

$$a : b = c : d,$$

it states

$$M_n(b) < M_m(a) \iff M_n(d) < M_m(c).$$

In V.13 the implication is used from right to left, since the strict witness is initially attached to $c : d$.

13.7 Lean Proof Architecture

The proof DAG is

$$\begin{array}{c}
 \text{hGreater} : c : d > e : f \\
 \Downarrow \\
 \exists m, n, M_n(d) < M_m(c) \wedge \neg(M_n(f) < M_m(e)) \\
 \Downarrow \\
 \text{hProp} : a : b = c : d \\
 \Downarrow \\
 M_n(b) < M_m(a) \iff M_n(d) < M_m(c) \\
 \Downarrow \\
 M_n(b) < M_m(a) \\
 \Downarrow \\
 M_n(b) < M_m(a) \wedge \neg(M_n(f) < M_m(e)) \\
 \Downarrow \\
 a : b > e : f.
 \end{array}$$

No case split and no order trichotomy is needed.

13.8 Axiomatic Strength

The theorem requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

through the positive segment-class infrastructure.

In particular, V.13 does not use

```
HilbertArchimedeanPlane
```

and does not invoke the least-exceeding-multiple theorem.

The Archimedean work needed to create a greater-ratio witness belongs to V.8. V.13 merely transports a witness that has already been supplied.

13.9 Relationship with V.8

V.8 produces strict ratio comparisons from strict magnitude comparisons. For example,

$$a < c$$

can be turned into a comparison of ratios sharing a common second term.

V.13 then allows that strict ratio comparison to pass through an equality

$$a : b = c : d.$$

This is the middle step required in the strict part of V.14.

Schematically,

$$\begin{array}{c} a < c \\ \Downarrow \text{V.8} \\ c : d > a : d \\ a : b = c : d \\ \Downarrow \text{V.13} \\ a : b > a : d. \end{array}$$

V.10 then reads the final strict ratio comparison back as

$$b < d.$$

13.10 Relationship with V.10 and V.14

V.13 by itself does not produce an inequality between magnitudes. Its conclusion is still a strict comparison of ratios.

The transition back to magnitude order is V.10:

$$a : b > a : d \implies b < d.$$

Thus the strict-comparison mechanism of V.14 factors naturally as

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10}.$$

This factorization is explicit in `Proposition5_14.lean`.

13.11 Hidden Formal Data

The proof has one important hidden datum: the witness indices m, n .

Euclid's prose treats the strict ratio comparison abstractly. In the Lean definition, however, the hypothesis

```
HilbertEudoxusGreaterRatio Geo c d e f
```

contains explicit existential data.

The proof preserves that exact pair of indices. There is no second existence argument and no need to synchronize different witnesses.

This is why V.13 is so short formally.

13.12 Classical DAG versus Formal DAG

At the ratio level, the proposition is simply:

$$\begin{array}{c} a : b = c : d \\ c : d > e : f \\ \Downarrow \\ a : b > e : f. \end{array}$$

At the witness level:

$$\begin{array}{c} M_n(d) < M_m(c) \\ \neg(M_n(f) < M_m(e)) \\ \Downarrow \\ M_n(b) < M_m(a) \iff M_n(d) < M_m(c) \\ \Downarrow \\ M_n(b) < M_m(a) \\ \neg(M_n(f) < M_m(e)) \\ \Downarrow \\ a : b > e : f. \end{array}$$

The formal DAG therefore exposes the single comparison transported through Definition 5.

13.13 Structural Lesson

V.13 illustrates the value of separating equality of ratios from strict comparison of ratios.

Definition 5 supplies a family of comparison equivalences. Definition 7 supplies one witness pair m, n . V.13 combines them without any new magnitude construction.

This division of responsibilities is important for the later propositions: V.8 is responsible for existence, V.13 for transport, and V.10 for order reflection.

13.14 Conclusion

Proposition V.13 is the transport rule for strict ratio comparison through Eudoxean equality of ratios.

Its proof reuses the same Definition 7 witness and transports only the positive comparison clause through Definition 5. The negative clause is unchanged.

Although formally small, V.13 is the essential middle node of the proof graph

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10} \longrightarrow \text{V.14},$$

and therefore an important structural component of the Book V layer.

Chapter 14

Proposition V.14

14.1 Introduction

Proposition V.14 is the central comparison theorem of the present Book V reconstruction.

Given equal ratios

$$a : b = c : d,$$

it states that the comparison between the first terms a, c agrees with the comparison between the second terms b, d . Thus strict inequality, equality, and reverse strict inequality are all transported across the proportion.

In the formal development V.14 is not treated as an isolated ratio identity. Its strict direction is built from the already reconstructed propositions V.8, V.13, and V.10. The equality direction is then obtained by trichotomy and incompatibility between ratio equality and strict greater-ratio comparison.

The production file is

```
CGJteamLab/Proposition5_14.lean
```

and it imports

```
CGJteamLab.Proposition5_8  
CGJteamLab.Proposition5_10  
CGJteamLab.Proposition5_13
```

so that the proposition-level dependency graph is explicit at the module level.

14.2 Formal Statement of Proposition V.14

Assume

$$a : b = c : d.$$

The production theorem packages three equivalences:

$$a < c \iff b < d,$$

$$a = c \iff b = d,$$

and

$$c < a \iff d < b.$$

These are the three possible comparisons supplied by trichotomy on positive segment classes.

The production theorem is therefore stronger than a single implication: it records complete comparison preservation between corresponding terms of equal ratios.

14.3 Why V.14 Is Structurally Important

V.14 is the theorem that turns ratio equality into order information about individual magnitudes.

This is exactly the kind of step needed later in V.16. There, after scaling two equal ratios by arbitrary positive multiples, V.14 transfers the comparison between the scaled first terms to the scaled second terms.

In the classical route to XI.23, V.16 will then allow an inequality such as

$$OP < OL$$

to be transported to the corresponding inequality

$$PQ < LM.$$

Thus V.14 is not merely an internal Book V convenience. It is one of the main order-transfer engines of the later geometry.

14.4 Strict Direction: $a < c \Rightarrow b < d$

The core proposition-local theorem is

```
hilbertEudoxus_v14_lt_second_of_lt_first
```

with hypotheses

$$a : b = c : d$$

and

$$a < c.$$

The proof has exactly three Euclid-numbered steps.

14.4.1 Step 1: V.8

From

$$a < c$$

V.8 gives, with the common second term d ,

$$c : d > a : d.$$

In Lean this is

```
hilbertEudoxus_v8_greater_first_same_second
```

applied to c, a, d .

14.4.2 Step 2: V.13

The given equality

$$a : b = c : d$$

transports the strict comparison

$$c : d > a : d$$

to

$$a : b > a : d.$$

This is exactly Proposition V.13.

14.4.3 Step 3: V.10

Now the two ratios

$$a : b \quad \text{and} \quad a : d$$

have the same first term. V.10 reflects their strict comparison back to the second magnitudes:

$$a : b > a : d \implies b < d.$$

Hence

$$a < c \implies b < d.$$

14.5 Strict-Direction Dependency DAG

The formal and classical proposition-level graph coincide:

$$\begin{aligned} & a < c \\ & \Downarrow \text{V.8} \\ & c : d > a : d \\ & a : b = c : d \\ & \Downarrow \text{V.13} \\ & a : b > a : d \\ & \Downarrow \text{V.10} \\ & b < d. \end{aligned}$$

This is the most important dependency chain in the Book V reconstruction up to this point.

14.6 Equality Direction: $a = c \Rightarrow b = d$

The proposition-local theorem is

```
hilbertEudoxus_v14_eq_second_of_eq_first
```

Assume

$$a : b = c : d$$

and

$$a = c.$$

After substitution the proportion becomes

$$a : b = a : d.$$

Now apply trichotomy to b, d :

$$b < d, \quad b = d, \quad d < b.$$

The middle case is the desired conclusion.

14.6.1 Case $b < d$

V.8 gives

$$a : b > a : d.$$

But we also have

$$a : b = a : d.$$

The interface theorem

```
hilbertEudoxusProportion_not_greater
```

states that Eudoxus-equal ratios cannot simultaneously stand in the greater-ratio relation.

Hence $b < d$ is impossible.

14.6.2 Case $d < b$

V.8 gives

$$a : d > a : b.$$

The original equality of ratios is first reversed using

```
hilbertEudoxusProportion_symm
```

to obtain

$$a : d = a : b.$$

Again equality of ratios excludes a simultaneous strict greater-ratio comparison.

Thus $d < b$ is impossible.

Only

$$b = d$$

remains.

14.7 The Full Less-Than Equivalence

The theorem

hilbertEudoxus_v14_lt_iff

states

$$a < c \iff b < d.$$

The forward direction is exactly the strict theorem already proved.

For the reverse direction assume

$$b < d$$

and apply trichotomy to a, c .

If

$$a < c,$$

we are done.

If

$$a = c,$$

the equality direction gives

$$b = d,$$

contradicting $b < d$.

If

$$c < a,$$

symmetry of the original proportion gives

$$c : d = a : b.$$

Applying the already established strict direction to $c < a$ then yields

$$d < b,$$

contradicting $b < d$.

Thus only

$$a < c$$

is possible.

14.8 The Full Equality Equivalence

The theorem

```
hilbertEudoxus_v14_eq_iff
```

states

$$a = c \iff b = d.$$

The forward direction is the equality theorem already proved.

The reverse direction is more delicate. Assume

$$b = d$$

and apply trichotomy to a, c .

If

$$a < c,$$

then the less-than equivalence gives

$$b < d,$$

contradicting $b = d$.

If

$$a = c,$$

we are done.

If

$$c < a,$$

reverse the proportion and use the less-than equivalence to obtain

$$d < b,$$

again contradicting $b = d$.

Therefore

$$a = c.$$

14.9 A Formalization Correction in the Equality Proof

The reverse implication

$$b = d \implies a = c$$

exposed an important directionality issue during compilation.

An earlier attempted proof tried to reverse the proportion and then reuse the lemma

$$\text{equality of first terms} \implies \text{equality of second terms.}$$

But after reversing

$$a : b = c : d$$

to

$$c : d = a : b,$$

the hypothesis $b = d$ is still an equality of the second terms, not of the first terms. The lemma therefore cannot be applied in that direction.

The corrected proof uses trichotomy on a, c and eliminates both strict cases via the already proved less-than equivalence.

This correction is mathematically informative: V.14 equality is not obtained by a naive symmetry of the one-way equality lemma. The comparison theorem is needed.

14.10 Reverse Strict Comparison

The theorem

```
hilbertEudoxus_v14_gt_iff
```

states

$$c < a \iff d < b.$$

No new argument is required.

Reverse the proportion

$$a : b = c : d$$

to

$$c : d = a : b$$

and apply the less-than equivalence already established.

Thus the third comparison form is a direct consequence of symmetry.

14.11 Formal Statement

The final production theorem is:

```
theorem euclid_proposition_5_14
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a b c d : HilbertPositiveSegmentClass Geo)
  (hProp : HilbertEudoxusProportion Geo a b c d) :
  (HilbertPositiveSegmentLess Geo a c <=>
    HilbertPositiveSegmentLess Geo b d) /\
  (a = c <=> b = d) /\
  (HilbertPositiveSegmentLess Geo c a <=>
    HilbertPositiveSegmentLess Geo d b) := by

exact
  \langle hilbertEudoxus_v14_lt_iff
    Geo a b c d hProp,
  hilbertEudoxus_v14_eq_iff
    Geo a b c d hProp,
  hilbertEudoxus_v14_gt_iff
    Geo a b c d hProp\rangle
```

The final wrapper simply packages the three comparison theorems.

14.12 Lean Proof Architecture

The production file contains five proposition-local components:

```
hilbertEudoxus_v14_lt_second_of_lt_first
hilbertEudoxus_v14_eq_second_of_eq_first
hilbertEudoxus_v14_lt_iff
hilbertEudoxus_v14_eq_iff
hilbertEudoxus_v14_gt_iff
```

followed by

```
euclid_proposition_5_14
```

which packages them.

The architecture is intentionally layered. The difficult mathematical input is isolated in the first

strict direction; the other comparison forms are then derived by trichotomy and symmetry.

14.13 Axiomatic Strength

The proposition is stated under

```
[HilbertIncidence Geo]
[HilbertArchimedeanPlane Geo]
```

The Archimedean assumption enters indirectly through V.8.

V.13 and V.10 themselves require only the weaker congruence-level magnitude infrastructure, but V.14 depends on V.8 to generate the strict comparison that starts the proof.

Thus the proposition-level assumption records the strongest dependency in the actual DAG rather than suggesting that every local step is Archimedean.

14.14 Relationship with V.8, V.10, and V.13

V.14 is the first proposition in the current Book V reconstruction whose formal module imports other Euclid-numbered Book V proposition files:

```
import CGJteamLab.Proposition5_8
import CGJteamLab.Proposition5_10
import CGJteamLab.Proposition5_13
```

This is deliberate.

The dependency graph is not hidden inside a monolithic interface file. Instead, the module structure records the mathematical proposition graph:

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10} \longrightarrow \text{V.14}.$$

The actual data flow is:

$$a < c \xRightarrow{\text{V.8}} c : d > a : d \xRightarrow{\text{V.13}} a : b > a : d \xRightarrow{\text{V.10}} b < d.$$

14.15 Relationship with V.16

V.14 is the decisive order-transfer theorem used in V.16 alternando.

To prove

$$a : c = b : d$$

from

$$a : b = c : d,$$

V.16 fixes arbitrary positive-multiple indices m, n and first constructs an equality of ratios

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.14 then gives all three comparison equivalences:

$$M_m(a) < M_m(b) \iff M_n(c) < M_n(d),$$

$$M_m(a) = M_m(b) \iff M_n(c) = M_n(d),$$

and

$$M_n(c) < M_n(d) \iff M_m(a) < M_m(b).$$

Those three clauses are exactly Definition 5 for the alternated ratio.

Thus V.14 is the order-theoretic heart of the formal proof of V.16.

14.16 Connection with the XI.23 Program

The original motivation for formalizing this fragment of Book V came from Euclid XI.23.

The classical XI.23 argument reaches a proportion of the form

$$OL : OP = LM : PQ.$$

If

$$OP < OL,$$

then the comparison-preservation content supplied by the Book V layer gives the corresponding comparison

$$PQ < LM.$$

In the present architecture this kind of transfer is encoded by the same Eudoxean comparison machinery that culminates in V.14 and V.16.

Thus the Book V reconstruction is not detached infrastructure: it supplies the exact ratio-order mechanism needed by the later solid-geometry proof.

14.17 Hidden Formal Data

Several features invisible in the printed ratio notation become explicit in Lean.

14.17.1 Three comparison forms are separate theorems

Although mathematically they are parts of one trichotomy statement, the formal development proves the $<$, $=$, and reverse- $<$ forms in a controlled order so that later proofs can reuse the exact direction needed.

14.17.2 Symmetry of proportion is explicit

Switching

$$a : b = c : d$$

to

$$c : d = a : b$$

is performed by

```
hilbertEudoxusProportion_symm
```

rather than being treated as syntactic interchange.

14.17.3 Strict and equal ratio relations are incompatible

The equality case relies on the reusable theorem

```
hilbertEudoxusProportion_not_greater
```

which formalizes the fact that ratios equal in the sense of Definition 5 cannot simultaneously satisfy Definition 7 in the same direction.

14.18 Classical DAG versus Formal DAG

The central strict direction has the proposition graph

$$\begin{array}{c}
 a < c \\
 \Downarrow \text{V.8} \\
 c : d > a : d \\
 \Downarrow \text{V.13} \quad \text{using } a : b = c : d \\
 a : b > a : d \\
 \Downarrow \text{V.10} \\
 b < d.
 \end{array}$$

The complete formal proposition then adds:

$$\begin{array}{c}
a = c \\
\Downarrow \\
\text{trichotomy on } b, d \\
\Downarrow \\
b = d,
\end{array}$$

and uses symmetry plus trichotomy to derive the converse and reverse strict forms.

At module level the same graph appears through imports:

```

Proposition5_8
Proposition5_10
Proposition5_13
  ↓
Proposition5_14.

```

14.19 Structural Lesson

V.14 shows why preserving the proposition-level DAG matters.

One could hide V.8, V.10, and V.13 as anonymous helpers inside a large Book V interface and prove V.14 from them. The final theorem would compile, but the Euclidean structure would disappear.

The current organization keeps two layers distinct:

- `HilbertInterfaceV.lean` contains reusable magnitude and Eudoxean infrastructure;
- `Proposition5_n.lean` files expose Euclid’s numbered theorem graph.

V.14 is the first place where this separation becomes visibly valuable: the Lean import graph mirrors the mathematical proof graph.

14.20 Conclusion

Proposition V.14 is the comparison-preservation theorem for equal Eudoxean ratios.

Its strict core is the three-step chain

$$V.8 \longrightarrow V.13 \longrightarrow V.10,$$

while equality and reverse strict comparison are derived by trichotomy and symmetry.

The proposition packages the complete statement

$$a < c \iff b < d, \quad a = c \iff b = d, \quad c < a \iff d < b,$$

and is the decisive order-transfer tool later used in V.16 alternando and, through that theorem, in the classical Book XI program.

Chapter 15

Proposition V.15

15.1 Introduction

Proposition V.15 records the invariance of a ratio under taking the same positive multiple of both terms.

In the present reconstruction the proposition is formulated for the positive multiple operation already defined in the Book V interface. If $M_k(a)$ denotes the $(k + 1)$ -fold positive multiple of a , then the theorem states

$$a : b = M_k(a) : M_k(b).$$

The proof is direct from Eudoxus' Definition 5. No Archimedean argument is needed, and no strict greater-ratio witness is constructed.

The production file is

```
CGJteamLab/Proposition5_15.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the shared Book V magnitude infrastructure.

15.2 Formal Statement

The production theorem is:

```
theorem euclid_proposition_5_15
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b : HilbertPositiveSegmentClass Geo)
  (k : Nat) :
  HilbertEudoxusProportion
```

```

Geo
a b
(hilbertPositiveSegmentMultiple Geo k a)
(hilbertPositiveSegmentMultiple Geo k b) := by

intro m n

constructor
case left =>
  exact
    (hilbertPositiveSegmentMultiple_nested_lt_iff
      Geo m n k a b).symm

case right =>
  constructor
  case left =>
    exact
      (hilbertPositiveSegmentMultiple_nested_eq_iff
        Geo m n k a b).symm
  case right =>
    exact
      (hilbertPositiveSegmentMultiple_nested_lt_iff
        Geo n m k b a).symm

```

The code shown here is an ASCII presentation of the same proof structure as the production theorem.

15.3 Positive Multiples

The operation

```

hilbertPositiveSegmentMultiple Geo k a

```

uses shifted indexing:

$$M_0(a) = a, \quad M_1(a) = a + a, \quad M_2(a) = a + a + a, \quad \dots$$

Thus every natural-number index represents a positive multiple.

For Proposition V.15, the same fixed index k is applied to both terms of the ratio.

15.4 What Definition 5 Requires

To prove

$$a : b = M_k(a) : M_k(b),$$

Definition 5 requires that for every pair of positive-multiple indices m, n , the following three comparisons agree:

$$M_m(a) < M_n(b),$$

$$M_m(a) = M_n(b),$$

and

$$M_n(b) < M_m(a),$$

with the corresponding comparisons after applying the common outer multiple k .

Thus the formal target reduces to three nested-multiple equivalences.

15.5 Nested Multiples

The reusable infrastructure proves:

```
hilbertPositiveSegmentMultiple_nested_lt_iff
hilbertPositiveSegmentMultiple_nested_eq_iff
```

Conceptually these state that applying the same positive outer multiple to both sides preserves and reflects strict order and equality.

For example,

$$M_m(M_k(a)) < M_n(M_k(b)) \iff M_m(a) < M_n(b),$$

and similarly for equality.

The reverse strict comparison is obtained by applying the same theorem with the two magnitudes and indices exchanged.

15.6 Why No Arithmetic of Scalar Multiplication Is Needed

One might expect a proof based on the numerical identity

$$(m + 1)(k + 1) = (k + 1)(m + 1).$$

The formal development avoids introducing scalar multiplication of magnitudes by natural numbers as a separate algebraic structure.

Instead, the required commutation of iterated positive addition is already isolated in the interface theorem

```
hilbertPositiveSegmentMultiple_commute
```

and the nested order/equality lemmas are built on top of it.

Thus V.15 remains internal to the synthetic additive magnitude layer.

15.7 Lean Proof Architecture

The complete proof has the form

$$\begin{array}{c}
 \text{fix arbitrary } m, n \\
 \Downarrow \\
 M_m(a) < M_n(b) \iff M_m(M_k(a)) < M_n(M_k(b)) \\
 \Downarrow \\
 M_m(a) = M_n(b) \iff M_m(M_k(a)) = M_n(M_k(b)) \\
 \Downarrow \\
 M_n(b) < M_m(a) \iff M_n(M_k(b)) < M_m(M_k(a)) \\
 \Downarrow \\
 a : b = M_k(a) : M_k(b).
 \end{array}$$

There are no proposition-local helper lemmas.

15.8 Axiomatic Strength

The proposition requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

through the positive segment-class infrastructure.

In particular, it does not require

```
HilbertArchimedeanPlane
```

and therefore does not depend on the Archimedean step used in V.8.

This is structurally natural: V.15 does not assert the existence of a sufficiently large multiple. It only compares already specified positive multiples.

15.9 Relationship with Definition 5

V.15 is almost a normalization theorem for Definition 5.

The equality of ratios is invariant under applying the same positive multiple to both terms because all three comparison types are invariant under that operation.

The proposition therefore demonstrates that the Eudoxean definition is compatible with the additive multiple structure developed earlier in `HilbertInterfaceV.lean`.

15.10 Relationship with V.16

V.15 is used twice in V.16.

Fix arbitrary indices m, n . From V.15 we get

$$a : b = M_m(a) : M_m(b)$$

and

$$c : d = M_n(c) : M_n(d).$$

Starting from

$$a : b = c : d,$$

V.11 composes these equalities to obtain

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

Then V.14 transfers the comparison of the first terms to the second terms.

Thus the proposition-level dependency chain is

$$\text{V.15} \longrightarrow \text{V.11} \longrightarrow \text{V.14} \longrightarrow \text{V.16}.$$

15.11 Hidden Formal Data

The main formal datum hidden in the classical statement is the indexing convention.

Because the project uses positive multiples only, the natural index k denotes $k + 1$ copies. Consequently the theorem is valid uniformly for all

$$k : \mathbb{N},$$

including $k = 0$, where it becomes the reflexive proportion

$$a : b = a : b.$$

No separate positivity proof for the multiplier is required.

15.12 Classical DAG versus Formal DAG

At the mathematical level:

$$\begin{array}{c}
a : b \\
\Downarrow \text{ common positive multiple } k \\
M_k(a) : M_k(b) \\
\Downarrow \\
a : b = M_k(a) : M_k(b).
\end{array}$$

At the Lean level:

$$\begin{array}{c}
\text{nested_lt_iff} \\
\text{nested_eq_iff} \\
\text{nested_lt_iff with reversed arguments} \\
\Downarrow \\
\text{three clauses of Definition 5} \\
\Downarrow \\
\text{euclid_proposition_5_15.}
\end{array}$$

The formal proof is therefore a direct structural reading of the definition.

15.13 Structural Lesson

V.15 shows that once positive multiples have been given a clean reusable interface, a classical proposition about equimultiples can become very short.

The mathematical work lies below the Euclid-numbered theorem: commutation of iterated addition and preservation/reflection of order and equality under a common positive multiple.

Keeping those facts in `HilbertInterfaceV.lean` leaves `Proposition5_15.lean` as the correct proposition-level statement rather than as a second arithmetic library.

15.14 Conclusion

Proposition V.15 establishes

$$a : b = M_k(a) : M_k(b)$$

for every positive-multiple index k .

Its proof is a direct verification of Eudoxus' Definition 5 using the nested multiple order and equality equivalences already available in the shared Book V interface.

The proposition is formally small but strategically important: it supplies the scaling step used twice in the proof of V.16 alternando.

Chapter 16

Proposition V.16

16.1 Introduction

Proposition V.16 is the alternando theorem.

From

$$a : b = c : d,$$

it derives

$$a : c = b : d.$$

This is the proposition that originally motivated the present Book V reconstruction. In the classical proof of Euclid XI.23, alternando is used to transform a proportion obtained from similar triangles into a form from which an inequality of corresponding magnitudes can be read.

The present formal proof preserves the Euclidean proposition-level dependency graph. It uses V.15 to pass to arbitrary positive multiples, V.11 to compose equal ratios, and V.14 to transfer comparison across those scaled equal ratios.

The production file is

```
CGJteamLab/Proposition5_16.lean
```

and it imports

```
CGJteamLab.Proposition5_11
CGJteamLab.Proposition5_14
CGJteamLab.Proposition5_15
```

so that the mathematical dependency graph is explicit in the module structure.

16.2 Euclid's Alternando Statement

The starting hypothesis is

$$a : b = c : d.$$

The desired conclusion is

$$a : c = b : d.$$

To verify the conclusion by Definition 5, we must fix arbitrary positive multiple indices m, n and prove that the three comparisons between

$$M_m(a) \quad \text{and} \quad M_n(c)$$

agree with the corresponding comparisons between

$$M_m(b) \quad \text{and} \quad M_n(d).$$

This is exactly the form supplied by V.14 once an appropriate scaled proportion has been constructed.

16.3 Step 1: Scale the First Ratio by V.15

For arbitrary m, n , V.15 gives

$$a : b = M_m(a) : M_m(b).$$

In Lean:

```
have hScaleAB :
  HilbertEudoxusProportion
    Geo
    a b
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b) :=
  euclid_proposition_5_15
  Geo a b m
```

This proportion is then reversed so that the scaled ratio appears first.

16.4 Step 2: Scale the Second Ratio by V.15

Similarly,

$$c : d = M_n(c) : M_n(d).$$

In Lean:

```

have hScaleCD :
  HilbertEudoxusProportion
    Geo
    c d
    (hilbertPositiveSegmentMultiple Geo n c)
    (hilbertPositiveSegmentMultiple Geo n d) :=
  euclid_proposition_5_15
  Geo c d n

```

The two arbitrary indices m, n are independent, exactly as required by Definition 5 for the alternated proportion.

16.5 Step 3: Compose Ratios by V.11

The original proportion is

$$a : b = c : d.$$

After reversing the first V.15 equality we have

$$M_m(a) : M_m(b) = a : b.$$

V.11 composes this with the original proportion to obtain

$$M_m(a) : M_m(b) = c : d.$$

A second use of V.11 with

$$c : d = M_n(c) : M_n(d)$$

gives the fully scaled equality

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

This is the central bridge of the proof.

16.6 Step 4: Apply V.14

Now V.14 applies to the scaled equality

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

It yields the three comparison equivalences

$$M_m(a) < M_n(c) \iff M_m(b) < M_n(d),$$

$$M_m(a) = M_n(c) \iff M_m(b) = M_n(d),$$

and

$$M_n(c) < M_m(a) \iff M_n(d) < M_m(b).$$

But these are exactly the three clauses of Definition 5 for

$$a : c = b : d.$$

Since m, n were arbitrary, alternando follows.

16.7 Proposition-Level Dependency DAG

The complete Euclid-numbered graph is

$$\begin{array}{ccc}
 & \text{V.15} & \text{V.15} \\
 & \Downarrow & \Downarrow \\
 a : b = M_m(a) : M_m(b) & & c : d = M_n(c) : M_n(d) \\
 & \Downarrow & \\
 & \text{V.11} & \\
 & \Downarrow & \\
 M_m(a) : M_m(b) = M_n(c) : M_n(d) & & \\
 & \Downarrow & \\
 & \text{V.14} & \\
 & \Downarrow & \\
 \text{three comparison equivalences} & & \\
 & \Downarrow & \\
 & \text{V.16.} &
 \end{array}$$

The original hypothesis

$$a : b = c : d$$

enters at the V.11 composition stage.

16.8 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_16
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a b c d : HilbertPositiveSegmentClass Geo)
  (hProp :
    HilbertEudoxusProportion Geo a b c d) :
    HilbertEudoxusProportion Geo a c b d := by

intro m n

have hScaleAB :=
  euclid_proposition_5_15
  Geo a b m

have hScaleCD :=
  euclid_proposition_5_15
  Geo c d n

have hScaledAB_to_CD :=
  euclid_proposition_5_11
  Geo
  (hilbertPositiveSegmentMultiple Geo m a)
  (hilbertPositiveSegmentMultiple Geo m b)
  a b
  c d
  (hilbertEudoxusProportion_symm
    Geo
    a b
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b)
    hScaleAB)
  hProp

have hScaled :=
  euclid_proposition_5_11
  Geo
  (hilbertPositiveSegmentMultiple Geo m a)
  (hilbertPositiveSegmentMultiple Geo m b)
  c d
  (hilbertPositiveSegmentMultiple Geo n c)
  (hilbertPositiveSegmentMultiple Geo n d)
  hScaledAB_to_CD
  hScaleCD

constructor
case left =>
  exact
    hilbertEudoxus_v14_lt_iff
    Geo
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b)
    (hilbertPositiveSegmentMultiple Geo n c)
    (hilbertPositiveSegmentMultiple Geo n d)
    hScaled

case right =>
  constructor

```

```

case left =>
  exact
    hilbertEudoxus_v14_eq_iff
      Geo
      (hilbertPositiveSegmentMultiple Geo m a)
      (hilbertPositiveSegmentMultiple Geo m b)
      (hilbertPositiveSegmentMultiple Geo n c)
      (hilbertPositiveSegmentMultiple Geo n d)
      hScaled
case right =>
  exact
    hilbertEudoxus_v14_gt_iff
      Geo
      (hilbertPositiveSegmentMultiple Geo m a)
      (hilbertPositiveSegmentMultiple Geo m b)
      (hilbertPositiveSegmentMultiple Geo n c)
      (hilbertPositiveSegmentMultiple Geo n d)
      hScaled

```

The displayed proof uses only ASCII syntax in the documentation; it is mathematically identical to the production proof.

16.9 Why V.16 Needs V.14

A tempting approach is to try to prove *alternando* directly from the original Definition 5 hypothesis

$$a : b = c : d.$$

But Definition 5 compares multiples of a with multiples of b , and multiples of c with multiples of d .

Alternando instead requires comparison of

$$M_m(a) \text{ with } M_n(c),$$

and simultaneously

$$M_m(b) \text{ with } M_n(d).$$

This is a cross-comparison between the two original ratios.

V.14 is precisely the theorem that converts an equality of ratios into such cross-comparison information.

16.10 Why V.15 Is the Correct Scaling Step

V.14 can compare corresponding terms only after we have constructed the scaled proportion

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.15 supplies the two links

$$a : b = M_m(a) : M_m(b),$$

and

$$c : d = M_n(c) : M_n(d).$$

V.11 then composes these links with the original equality.

Thus the proof is not merely an algebraic rearrangement of symbols. It reconstructs the exact Eudoxean comparison environment needed for arbitrary m, n .

16.11 Axiomatic Strength

The theorem is stated under

```
[HilbertIncidence Geo]
[HilbertArchimedeanPlane Geo]
```

The Archimedean assumption enters through V.14, whose strict comparison component depends on V.8.

V.11 and V.15 themselves require only the weaker congruence-level magnitude infrastructure.

The assumption profile therefore reflects the deepest dependency in the proposition graph.

16.12 Relationship with V.11

V.11 appears twice in the proof.

First:

$$M_m(a) : M_m(b) = a : b$$

is composed with

$$a : b = c : d$$

to obtain

$$M_m(a) : M_m(b) = c : d.$$

Second, that result is composed with

$$c : d = M_n(c) : M_n(d)$$

to obtain

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

Thus V.11 is the glue connecting the two V.15 scaling results to the original proportion.

16.13 Relationship with V.14

Once the fully scaled proportion has been constructed, V.14 finishes the mathematical argument.

No additional order lemma is needed in V.16 itself. The three comparison equivalences returned by V.14 are already exactly the three components of the target Definition 5 statement.

This is an important architectural point: the order theory remains localized inside V.14.

16.14 Connection with Euclid XI.23

The motivation for formalizing V.16 came from the classical proof of XI.23.

That proof reaches a proportion of the form

$$OL : LM = OP : PQ.$$

V.16 transforms it to

$$OL : OP = LM : PQ.$$

Now a strict comparison such as

$$OP < OL$$

can be transferred to the corresponding comparison

$$PQ < LM.$$

This is the ratio-order step that the earlier proposition-specific XI.23 experiments were reproducing indirectly.

The Book V reconstruction isolates it as reusable mathematics.

16.15 Two Ratio Languages in the Project

The project already contains another proportion language:

```
HilbertPositiveSegmentProportion
```

based on an angle-coded ratio witness.

V.16 in the present file is instead proved for

```
HilbertEudoxusProportion
```

defined by comparison of all positive multiples.

These two notions are not identified by definition and are not silently interchanged.

A future bridge theorem may establish their equivalence under suitable hypotheses, but V.16 itself belongs entirely to the Eudoxean layer.

16.16 Hidden Formal Data

Several objects implicit in the classical notation become explicit.

16.16.1 Arbitrary multiple indices

The proof begins with

```
intro m n
```

because the target proportion is Definition 5.

Thus V.16 is proved pointwise for arbitrary comparison indices.

16.16.2 Two independent scalings

The first ratio is scaled by m , the second by n . They are not the same multiplier.

This independence is essential: Definition 5 compares arbitrary multiples of the first and third magnitudes.

16.16.3 Symmetry is explicit

The V.15 statement gives

$$a : b = M_m(a) : M_m(b),$$

whereas the first V.11 composition needs

$$M_m(a) : M_m(b) = a : b.$$

The reversal is therefore performed explicitly by

```
hilbertEudoxusProportion_symm
```

rather than by informal rewriting.

16.17 Classical DAG versus Formal DAG

The mathematical core is:

$$\begin{array}{c}
 a : b = c : d \\
 \Downarrow \\
 a : b = M_m(a) : M_m(b) \quad c : d = M_n(c) : M_n(d) \\
 \Downarrow \\
 M_m(a) : M_m(b) = M_n(c) : M_n(d) \\
 \Downarrow \\
 \text{V.14 comparison transfer} \\
 \Downarrow \\
 M_m(a) ? M_n(c) \iff M_m(b) ? M_n(d) \\
 \Downarrow \\
 a : c = b : d.
 \end{array}$$

The Lean module graph is:

```

Proposition5_15
Proposition5_11
Proposition5_14
  \Downarrow
Proposition5_16.

```

This is a good example of the project principle that Euclid-numbered propositions should remain visible as separate nodes of the formal dependency graph.

16.18 Structural Lesson

V.16 vindicates the decision to build a genuine Book V layer rather than to continue adding local ratio helpers inside XI.23.

The theorem is not difficult once the correct infrastructure exists, but its proof draws on several distinct reusable ideas:

- common positive multiples;
- transitivity of ratio equality;
- preservation of comparison across equal ratios;
- Eudoxus' Definition 5.

By separating these into V.15, V.11, V.14, and the shared interface, the formal proof mirrors the mathematical architecture rather than hiding it in one large proposition-specific argument.

16.19 Conclusion

Proposition V.16 establishes *alternando*:

$$a : b = c : d \implies a : c = b : d.$$

For arbitrary positive-multiple indices m, n , V.15 scales the two original ratios, V.11 composes the resulting equalities, and V.14 transfers the three possible comparisons between the scaled first terms to the scaled second terms.

Those three equivalences are exactly Definition 5 for the alternated ratio.

This completes the present Book V fragment needed for the classical route back to Euclid XI.23.

Bibliography

- [1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.